Reproducing LucidPPN: Interpretable Bird Classification with Separate Shape and Colour Evidence

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Paper: Pach, Lewandowska, Tabor, Zieliński, Rymarczyk. *LucidPPN: Unambiguous Prototypical Parts Network for User-centric Interpretable Computer Vision.* ICLR 2025.

Code: https://github.com/rayyansgit/lucidppn-reproduction. Branch `main`: reproduction (Milestone 2). Branch `m3&m4`: experiment and deployment (Milestones 3-4).

1. Introduction and problem statement

Image classifiers rarely explain their decisions. Prototypical-part networks do: they classify an image by matching its parts to parts of training images ("this part looks like that part of a Cardinal").

LucidPPN identifies two problems with these explanations:

1. **A match mixes shape, texture and colour.** The user cannot tell which one the model used.
2. **Prototypes of different classes sit on different body parts.** Two classes cannot be compared part by part.

LucidPPN fixes both by design. We reproduced the method on a subset of a bird dataset, tested one of its claims with our own experiment, and deployed the trained model as an application.

2. Background

- **ProtoPNet** (Chen et al., 2019): classification by comparing image patches with learned prototypes.
- **ProtoPShare, ProtoPool** (Rymarczyk et al., 2021, 2022): prototypes shared between classes.
- **PIP-Net** (Nauta et al., 2023): prototypes closer to human notions of a part; LucidPPN's main baseline.
- **Nauta et al. (2021)**: users often cannot tell whether a prototype matched on colour or shape.
- **PDiscoNet** (van der Klis et al., 2023): finds object parts using class labels only. LucidPPN uses its part masks.

3. Paper methodology

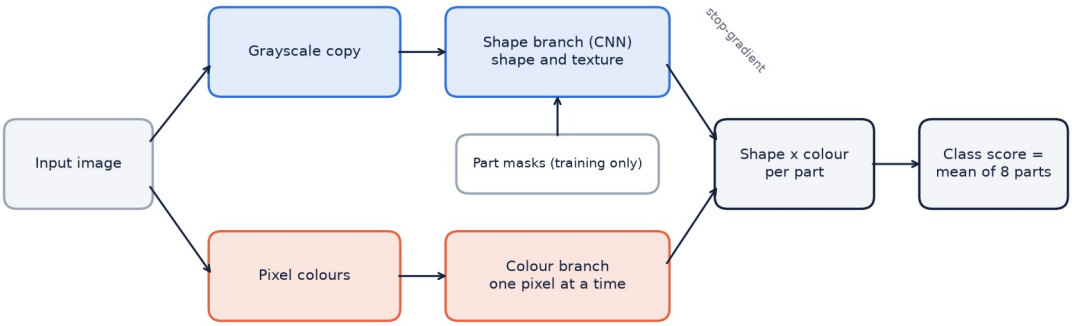


Figure 1. LucidPPN: shape and colour branches, combined per part.

- **Shape branch.** A CNN sees only a grayscale copy of the image. For every part k and class m it produces a map of how much each location looks like part k of class m.
- **Colour branch.** A small network applied to one pixel at a time, so it learns colour but never shape.
- **Fusion.** Per part, evidence = shape map × colour map (best location). The class score is the mean over parts. The shape map is passed with a stop-gradient, so colour can confirm or block shape evidence but never changes what the shape branch learns.
- **Part alignment.** A training loss pushes each part's shape map towards PDiscoNet's mask for that part, so prototype k means the same body part in every class.

Datasets: CUB-200-2011, Stanford Cars, Stanford Dogs, Oxford Flowers. **Metrics:** top-1 accuracy, overlap of prototype maps with part masks, user studies.

Results on CUB (paper):

	Accuracy
LucidPPN	81.5 ± 0.4%
PIP-Net (baseline)	84.3%
Shape branch alone	80.4%
Under random colour (hue) changes: LucidPPN / shape branch	81.9 → 71.7% / 80.3 → 80.3%

4. Implementation details

We forked the authors' code and kept their model, losses and training step unchanged, so differences come from scale and settings, not from re-implementation. We wrote the data preparation, training and evaluation scripts, the experiments and the deployment. `PROVENANCE.md` lists the origin of every file.

Pipeline: (1) prepare the subset, (2) train PDiscoNet and save 8 part masks per image, (3) train LucidPPN with those masks, (4) evaluate, (5) run the experiment, (6) deploy.

	Paper	Ours
Species	200	25
Shape backbone	ConvNeXt-Tiny	ResNet18 (same 28 × 28 feature maps)
Parts per bird	12	8
Part detector	448 px, 28 epochs	224 px, 5 epochs (28 in a later study)
Training	40 epochs, batch 64, learning rate 0.002	82 epochs, batch 32, learning rate 0.0005
Runs	3	12 models in total (3-4 per setting)

Process: a one-batch check, an 8-species trial, then 25 species. Every run logs train and test accuracy per epoch. Hardware: Google Colab (T4 GPU), then a laptop RTX 2080 for the multi-run study.

5. Dataset and preprocessing

CUB-200-2011: 11,788 photos of 200 North American bird species, with a bounding box and 15 marked body points (beak, eye, wing, tail, ...) per bird.

Our subset: the first 25 species by CUB's numbering.

Included	Black-footed, Laysan and Sooty Albatross; Groove-billed Ani; Crested, Least, Parakeet and Rhinoceros Auklet; Brewer's, Red-winged, Rusty and Yellow-headed Blackbird; Bobolink; Indigo, Lazuli and Painted Bunting; Cardinal; Spotted and Gray Catbird; Yellow-breasted Chat; Eastern Towhee; Chuck-will's-widow; Brandt's, Red-faced and Pelagic Cormorant
Images	750 training (30 per species), 652 test, using CUB's official split
Omitted	The other 175 species, including most of CUB's hardest look-alike groups (sparrows, warblers, gulls, terns, vireos, wrens)
Why	Training time on free GPUs

Preprocessing (as in the paper):

- Each photo is cropped to its bounding box.
- Training: resize to 232 px, random geometric changes and flips, random crop to 224 px. Colours are never altered, so the colour branch learns real colours.
- Testing: resize to 224 px.
- The shape branch receives a normalised grayscale copy; the colour branch receives the colour image.

Consequence: chance is 4% (not 0.5%), and the task is easier than the full dataset. Our accuracy cannot be compared directly with the paper's. Most of our species are grey, brown or black, which matters for the colour experiment (Section 7).

6. Reproduction results

We always report accuracy at the last epoch. Without a validation set, picking the best epoch would mean choosing on the test set.

Run	Change	Test accuracy
Jump 2	10 epochs, no part masks	33.9%
Jump 3	10 epochs, with part masks	41.6%
Jump 4	82 epochs, batch 32, longer warm-up	81.9%
Jump 4, 4 runs	same settings repeated	**83.1 ± 1.0%**
Paper	200 species, ConvNeXt-Tiny	81.5 ± 0.4%

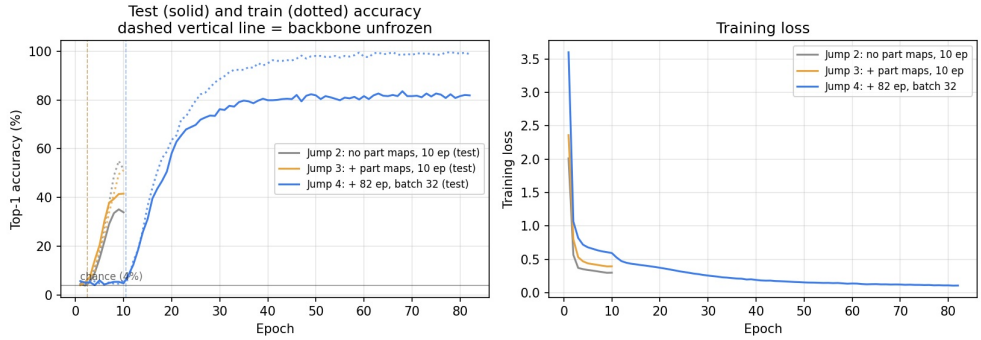


Figure 2. Test (solid) and train (dashed) accuracy per epoch, Jumps 2-4.

- **Part masks reduce overfitting:** +7.7 points at the same training accuracy; the train-test gap roughly halved.
- **Longer training gives most of the gain.** Accuracy levels off after about 40 epochs.
- **Run-to-run noise is about ±1-2 points.** Re-running Jump 4 with identical settings gave 84.0% instead of 81.9%.

Follow-up study (3 runs per setting).

	5-epoch part detector	28-epoch part detector
Our settings (Jump 4)	83.1 ± 1.0%	83.0 ± 0.7%
Paper's settings (authors' launch script)	81.0 ± 1.7%	81.4 ± 2.0%

- **The paper's settings score about 2 points lower here.** On 750 images, 40 epochs at batch 64 give 4× fewer training steps than ours.
- **A better part detector did not change accuracy, but improved where prototypes look.** Its parts match CUB's marked body points better (NMI 24.1 → 31.0), and prototypes trained with them land more consistently on the same body parts (17.6 → 20.8 and 18.1 → 22.6). This second score is PDiscoNet's metric applied to our prototypes; the LucidPPN paper does not report it, so we only use it to compare our own models.
- **Top-5 accuracy is 97-98%.** The most common errors are between look-alike dark species: Pelagic vs Brandt's Cormorant, Rusty vs Brewer's Blackbird, Black-footed vs Sooty Albatross (37% of all errors).

7. Additional experiment: ablation of the colour branch

Question: what does the colour branch add, and what does it cost when colours are misleading?

Method:

1. **Colour branch off.** Because of the stop-gradient, the shape branch never learns from colour. The shape branch inside a trained model is therefore the colour-free model, and no retraining is needed.
2. **Misleading colours.** Test images are recoloured with a random hue shift of up to 0.1, 0.25 or 0.5 of the colour wheel, keeping each pixel's brightness (the authors' method). The shape branch sees exactly the same image.
3. **Repeats.** All 12 trained models, 5 random recolourings each. Results are also split by how colourful each species is (Hasler and Süsstrunk colourfulness, measured on the test images).

Results (accuracy change in points, mean of 12 models; recoloured = hue shift up to 0.5):

Test images	Colour branch adds (original colours)	Full model when recoloured	Shape only when recoloured
All 25 species	+1.4 (positive in 12 of 12 models)	−4.1	0.0
12 most colourful species	+1.7	−7.5	0.0
12 least colourful species	+1.3	−0.9	0.0
Paper, 200 species	+1.4	−10.2	0.0

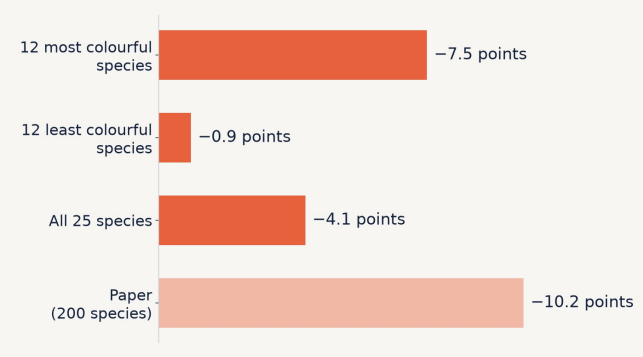


Figure 3. Accuracy lost by the full model when test images are recoloured (shape only: 0 in every group).

Findings:

- **Colour helps a little, consistently.** +1.4 points, the same as the paper. One model alone was not enough to show it (12 images fixed, 5 broken, $p \approx 0.14$); over 12 models it is clear ($p = 0.0005$).
- **The two branches are truly separate.** The shape-only model loses exactly 0 points in every model.
- **The cost falls on colourful birds.** On the most colourful species, misleading colours cost 7.5 points and push the full model below the shape-only model in 11 of 12 models. Grey species lose under 1 point. Cardinal loses 29 points.

Answer: the paper's trade-off holds. Colour adds about 1.4 points but makes the model vulnerable to wrong colours, and how vulnerable depends on how colourful the birds are. This is consistent with our smaller average loss (−4.1) than the paper's (−10.2), since most of our species are dull.

8. Analysis and discussion

- **Our accuracy only looks like the paper's.** Our model is weaker, but our task is much easier (25 species, few look-alikes). The similar numbers are a coincidence.
- **Part supervision serves the explanation more than accuracy.** Part masks reduced overfitting, but a better part detector changed where prototypes look, not accuracy. Interpretability needs its own measurements.
- **Several prototypes often share one spot.** In many images several of the 8 parts focus on the head, so they act as one piece of evidence. The app shows this.
- **Noise matters.** Two early conclusions changed once we had several runs: the first colour gain was not significant, and the slow start of training was mainly caused by the low default learning rate, not by the warm-up design.
- **Settings depend on dataset size.** The paper's settings, designed for 8× more images, did not transfer.

Limitations:

- Reduced scale: 25 species, ResNet18, 8 parts, weaker part detector.
- The first 25 species are mostly dull; a colourful subset would test the colour question better.
- No validation set; some settings were chosen after seeing test results.
- Only hue changes were tested; user studies, the other datasets and the PIP-Net comparison were not reproduced.

9. Deployment

A **Streamlit** web app and a **FastAPI** REST API, sharing one inference module and the trained model (84.0%). Both run on a laptop CPU; `run_demo.bat` / `run_demo.sh` install and start everything.

The app has three tabs:

1. **Prediction:** species, whether it is correct, top-3 scores.
2. **Explanation:** for each of the 8 parts, where the model looked, the matching part of a training bird (in grayscale, as the shape branch sees it), and whether the colour fits.
3. **Colour ablation:** full model and shape-only model side by side, with a slider that recolours the bird, plus the test-set results of Section 7.

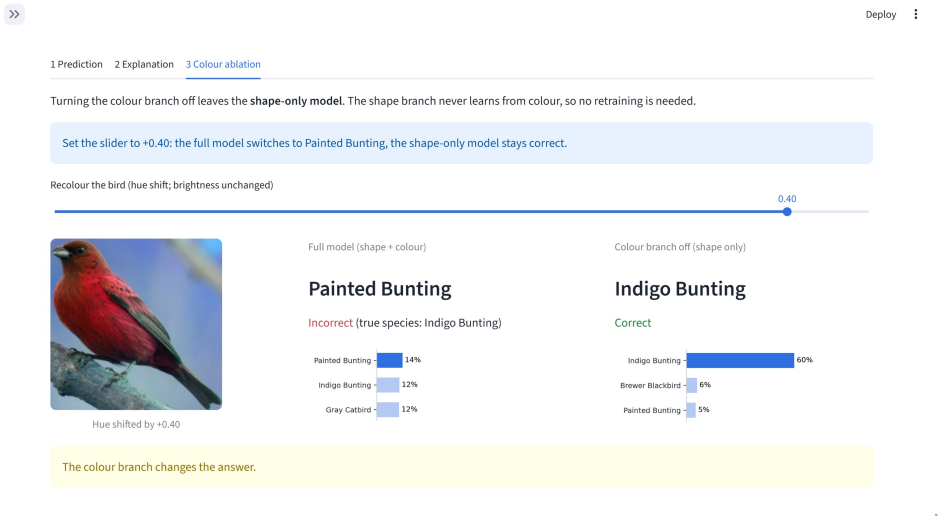


Figure 4. The app's colour-ablation tab: Indigo Bunting recoloured by +0.40.

In this example the recoloured Indigo Bunting is classified as a Painted Bunting by the full model, while the shape-only model stays correct: Section 7's result on a single image.

10. Conclusion

We reproduced LucidPPN's full pipeline on 25 CUB species (83.1 ± 1.0%). Our ablation confirmed the paper's colour trade-off: colour adds about 1.4 points, the shape branch is unaffected by recolouring, and the cost of misleading colours falls on colourful species. The deployed app shows the method's explanation part by part and runs the ablation live.

11. References

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